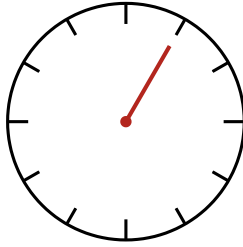


The Instrument



THE INSTRUMENT

*Twelve tools for a life
of your own making*

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THE INSTRUMENT

Twelve tools for a life of your own making

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This is copy number _____ of five hundred.

Signed _____

*For the one who will use it,
and the one it will be passed to.*

How we spend our days is, of course, how we spend our lives.
What we do with this hour, and that one, is what we are doing.

ANNIE DILLARD, *THE WRITING LIFE*, 1989

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Before you begin

Most books do not work. This is an odd thing to print at the front of one, but it is true, and it is the reason this book is shaped the way it is. A book rests on an unspoken theory of learning: that an author can think a thought, write it down, and a reader can absorb it by reading. The theory is wrong. Almost everyone who finishes a serious book retains very little of it a month later, and almost nobody notices. The words go in, feel like understanding, and go out again. You have probably read a hundred books that were supposed to change your life. Count the ones that did.

The ones that did have something in common. You did something with them. You copied a sentence out, you tried a method, you argued with a chapter for a week, you gave a copy to someone and had to explain it. The change never came from the reading. It came from the use.

So this is not a book to be read. It is an instrument to be used, and it has been built accordingly. It is short, so that it can be read again. It is divided into twelve, so that it can be practised, one instrument a month, for a year. It is printed on paper that takes ink, with pages that are meant to be written on. It is numbered, so that it will be kept. And it is expensive, so that it will not be ignored. That last point is not a joke. Things we pay dearly for, we use; the price is part of the mechanism.

The word *instrument* carries three meanings, and the book means all of them. An instrument is a tool, made for a purpose, that extends what a hand can do. An instrument is something you practise, badly at first, until it becomes an extension of you. And an instrument, in the oldest sense, is a document with binding force: a deed, a contract, a will. There is a page near the end of this book that asks for your signature. It is there because a book you have signed is harder to put down than one you have merely bought.

THE TWELVE

None of the twelve instruments is a secret, and none of them is new. Some are two thousand years old. Each has been chosen because it satisfies three tests: it changes what a person does, not only what they think; the change lasts; and there is evidence, from research or from long human practice, that

it works for people who are not unusual. Where an instrument leans on research, the research is named in the notes at the back, together with what it does not prove. Where it leans on a story, the story is labelled a story. The claim of this book is not that its ideas are original. It is that a year of using them is.

HOW TO USE IT

- 1 One instrument a month, in order. Read the chapter on the first day of the month. It takes twenty minutes. Do not read ahead. The instruments build on one another, and anticipation is part of the method.
- 2 Do the practice within the first seven days. By hand, with a pen, on the ledger page that faces it. Handwriting is slow, and slowness is the point: you cannot write a sentence by hand without deciding whether you mean it.
- 3 Carry the card for that instrument for the rest of the month. Put it where you will see it without looking for it: the desk, the mirror, the inside of the wallet. It is there to interrupt you.
- 4 On the last day of the month, return to the ledger page. Sign it, date it, and mark it *kept* or *revised*. Both are honourable. Blank is not.
- 5 After the twelfth month, sign the vow, and then open the sealed envelope that came with the book. Not before.
- 6 If you miss a month, you have not failed. Begin again on the next first day. The book does not mind. The book is patient in a way that you are not yet, and that is one of the things it is for.
- 7 Do not lend the book during the year. After the year, do. The colophon has room for the names of the people it belongs to next.

You will notice the book does not ask what you want. It asks what you will regret, what you attend to, what you would have to be true, whom you would knock for. These are stranger questions and they have better answers. Begin with the first one.

The set

This copy came in a box with five other things. None of them is decoration. Each has a job in the method, and each is named where it is needed.

1 THE BOOK

Ninety-six pages. Twelve instruments, each with a practice and a ledger page meant to be written on, in pen. The paper was chosen for it.

2 THE TWELVE CARDS

One card per instrument, printed letterpress, with the axiom on one face and the practice on the other. Carry the month's card where you will see it without looking for it.

3 THE WEEKS CHART

Four thousand six hundred and eighty boxes, one per week of a ninety-year life, with the eightieth year marked, folded like a map. It is filled in during the second month and consulted at every review after.

4 THE NEEDLE

A brass page-marker stamped with this copy's number. It marks the current month. When the year is finished, it marks the vow.

5 THE SEALED ENVELOPE

The thirteenth chapter. It is opened after the vow is signed, and not before. Owners who have opened it early report that it did not work, which is the truth.

6 THE REGISTER

Each copy can be registered with the publisher by its number. The register records the day your year began and sends one message on the first of each month, with nothing in it but the instrument's name. When the book is passed on, the next owner registers it again, and the year begins again.

INSTRUMENT ONE · DIRECTION



The Eulogy

Begin at the end.

Remembering that I'll be dead soon is the most important tool I've ever encountered to help me make the big choices in life.

STEVE JOBS, STANFORD, 2005

MONTH ONE

NOBODY, AT THE END, has asked to see the spreadsheet. The palliative nurse Bronnie Ware spent years sitting with people in their last weeks and wrote down what they said they regretted. The list, when she published it, went around the world, and it deserved to. They wished they had lived a life true to themselves rather than the life others expected. They wished they had not worked so hard. They wished they had said what they felt, had stayed in touch with their friends, had let themselves be happier. Not one of them mentioned a promotion.

The strange thing about this list is not that it is surprising. It is that it is completely unsurprising, and that almost nobody acts on it. We know how the story ends and we spend the middle chapters on the wrong things anyway. The reason, I think, is that *the end* is an abstraction, and abstractions do not move people. A eulogy is not an abstraction. It is a specific piece of writing, delivered by a specific person, about you, and it can be drafted this afternoon.

This is the first instrument because everything else in the book depends on it. The other eleven are ways of deciding, learning, attending and building. None of them can tell you what to build. This one can.

— WHY IT WORKS

When Daniel Pink collected regrets from more than sixteen thousand people in a hundred and five countries, they sorted themselves into four piles. Foundation regrets: I did not do the work, save the money, look after the body. Boldness regrets: I did not take the chance. Moral regrets: I did the wrong thing. Connection regrets: I let the relationship lapse. In our twenties, regrets of action and regrets of inaction arrive in roughly equal numbers. With age, inaction pulls ahead, until it outnumbers action about two to one. The things we did have a natural end. The things we did not do go on not having happened for the rest of our lives.

A eulogy works on this directly, and it works for a reason that will return several times in this book. Imagining an outcome as if it has already happened changes the quality of the thinking about it. Researchers call this prospective hindsight, and it is the same mechanism that makes a project

premortem more useful than a plan. Jeff Bezos used a version of it in 1994, deciding whether to leave a safe job to sell books on the internet. He projected himself to eighty and asked which choice the old man would regret. The old man did not care about the bonus.

You are not writing the eulogy to be sad. You are writing it to find out what you already think.

There is a second reason. People who can see their future self vividly treat that person better. In one experiment, adults shown an age-progressed image of their own face allocated more than twice as much of a windfall to retirement as those who saw their present face. A eulogy is an age-progressed image of a whole life, and it tends to change the allocation.

— HOW TO HOLD IT

The commonest way to fail at this exercise is to write the résumé instead. A eulogy is not about what you achieved. It is about what it was like to know you, and what was left in the people around you when you were gone. If your draft is a list of positions held, throw it away and choose a different speaker: someone who saw you on ordinary Tuesdays. The second way to fail is grandiosity. The speaker is not describing a saint. They are describing someone whose weeks added up to something, and the best test of a sentence is whether the speaker could say it without anyone in the room smiling at the exaggeration.

Out of the draft you will take three sentences. These are the governing lines, and they are not goals. A goal is a thing you can finish: run the marathon, ship the product. A governing line is a description of a person, in the past tense, that you cannot yet honestly claim: *she kept her body able to carry her to the people she loved; he told people what they meant to him while they could still hear it*. Goals live underneath governing lines. Any plan that satisfies a goal and violates a line is a bad plan, however attractive, and you will now be able to tell.

The story is told that in 1888 Alfred Nobel opened a French newspaper and read his own obituary, printed by mistake when his brother died, under the heading *the merchant of death is dead*. Nobody has ever found the newspaper, and the tale may be a legend. It survives because it describes the instrument exactly. A man who had spent his life making explosives was shown, while there was still time, the sentence that would be said about him, and he did not like it, and eight years later his will endowed the prizes that carry his name. It does not matter whether Nobel read the obituary. What matters is that you can.

Write it once, properly, this month. Then let the other eleven instruments act on it. You will be asked to read it again in a year, in a letter you have not opened yet.

PRACTICE · INSTRUMENT ONE

Write the eulogy

Ninety minutes · alone · by hand · within the first seven days of the month

- 01 ☐ **Choose the speaker.** Decide who is giving the eulogy: a child, a partner, an old friend, someone you worked beside for years, a stranger you once helped. Choose the person whose good opinion you would most want to have deserved. Not the one who would flatter you.
 - 02 ☐ **Set the scene in one line.** The year, the room, who is sitting in it. This is what makes the mechanism work: the outcome must feel as though it has already happened.
 - 03 ☐ **Write four hundred words in their voice.** Not what you achieved. What it was like to know you, what you did with your ordinary weeks, and what you left behind in people. Do not stop to edit. Do not stop before four hundred.
 - 04 ☐ **Read it aloud, once.** Mark every sentence that is not yet true. Those marks are the map of the coming year.
 - 05 ☐ **Extract three governing lines.** Choose the three marked sentences you most want to earn. Copy them onto the ledger page opposite, in the past tense, exactly as the speaker would say them. From today these three lines outrank every goal you have.
-

WHY IT WORKS Regrets of inaction grow for the rest of a life; a eulogy converts vague future regret into three specific present sentences, which is something a person can act on this week. The vividness is the mechanism, not a decoration.

WHERE IT FAILS If the draft took less than an hour, or reads like a profile, you wrote the résumé. Change the speaker to someone who saw you on ordinary days and write it again.

LEDGER · INSTRUMENT ONE

The governing lines

SPOKEN BY _____ IN THE YEAR _____

The first line.

The second line.

The third line.

One thing I will stop doing this month because of them.

One thing I will start.

SIGNED

REVIEWED ON

KEPT / REVISED

INSTRUMENT TWO · TIME

II

The Weeks

A life is four thousand weeks. Spend them on purpose.

It is not that we have a short time to live, but that we waste a lot of it.

SENECA, ON THE SHORTNESS OF LIFE, C. 49 AD

MONTH TWO

IF YOU LIVE TO EIGHTY, you will have had a little over four thousand weeks. Written as a number the figure is not alarming; it is smaller than the population of a village. Drawn as a grid, fifty-two boxes wide and eighty rows tall, it fits comfortably on a single sheet of paper, and the box for this week is smaller than a fingernail. Tim Urban drew that grid in 2014 and Oliver Burkeman built a book on the arithmetic, and the reaction of almost everyone who sees it for the first time is the same: a short silence, and then a change in the way they hold their calendar.

Seneca, who was rich, busy and eventually executed, made the same point two thousand years earlier. The problem is not that life is short. The problem is that we treat time as if it were a fluid, arriving in an endless stream, when it is in fact a countable set of boxes, most of which are already coloured in. Nobody wastes a thing they can count. We waste time because we have never been made to see it.

This chapter comes with a chart. It is folded in the box the book came in, and this month it will be filled in with a pen. The chart is not a memento mori, or not only that. It is an instrument of scale, and its purpose is allocation, not dread.

— WHY IT WORKS

The week is the right unit for a life. A day is too small to plan anything of consequence, and a year is too large to feel; but a week is the cycle at which the important things actually recur. Work recurs weekly. Rest recurs weekly. The people you love, if you see them at all, are seen on a weekly rhythm or not seen. A habit that survives its first twelve weeks usually survives. Whatever you do every week, you do with your life; whatever you do not do in a given week, you have not done, and the box is coloured in regardless.

Making time countable does two things. It makes it finite, and finite things get allocated. And it makes the future self, who will run out of boxes, into someone specific, whose interests you can weigh against your present one. Annie Dillard put the whole argument into one sentence: how we spend our days is, of course, how we spend our lives. The chart is that sentence, drawn.

The practice is old. Marcus Aurelius, who ruled an empire and wrote at night for nobody but himself, kept returning to it: you could leave life right now, he wrote; let that determine what you do and say and think. The Romans did not find this morbid. They found it clarifying, in the way that a deadline is clarifying, and the chart is simply the deadline made visible, one small box at a time. It does not ask you to think about death. It asks you to think about Thursday.

A week is the unit of a life: small enough to plan, large enough to matter, and it comes around again.

The most useful sum on the chart is not the total. It is what Urban called the tail end. Suppose your parents are seventy and you see them three times a year, for a weekend each. If they live to eighty-five, you will see them about forty-five more times. If you left home at eighteen and saw them daily until then, you have already used more than nine tenths of the time you will ever spend in their company. Run the same sum for the friend in another city, the child who will leave in six years, the sibling. The numbers are not cruel. They are simply true, and they are the fastest way to decide what a week is for.

— HOW TO HOLD IT

The chart produces two decisions. The first is a weekly non-negotiable: one block of at least two hours, at the same time every week, given to the thing your governing lines require and nothing else. It is not an aspiration; it is an appointment, and it goes in the calendar for the next twelve weeks before this chapter is finished. It will be interrupted. Defend it the way you would defend a flight you had paid for.

The second decision is a subtraction. Something recurs weekly in your life that you would not choose if it were offered fresh: a meeting, a standing obligation, a habit that fills the evening. Stop it. Not later, not after this month. The chart will show you why: a weekly hour is fifty hours a year, and

fifty hours a year for the rest of a working life is more than two thousand hours, which is more than a year of work.

Look, too, at the rows already coloured. They are not the wasted part. They are the base the rest is built on, and the chart makes an argument that is easy to miss when the days are lived one at a time: a good week repeated is not one good week, it is a row, and rows are what a life is made of.

Some owners keep the chart alive after the month is over, and it is worth the minute it costs. On Sunday evening, colour the week just finished and write a single word in the box: the place, the person, the thing that happened. After a year the row is a diary of the kind that nobody has the discipline to keep in any other form, and after ten it is the only honest record of a decade that most people will ever have.

Hold the chart lightly after that. It is a map, not a countdown. People who stare at it become anxious, and anxiety spends weeks too. Fill it, do the two sums, make the two decisions, and then fold it and put it away until the next month asks you to open it. The boxes will keep being coloured in whether you watch or not. The only question the chart asks is what colour.

PRACTICE · INSTRUMENT TWO

Fill the chart

One hour · the Weeks Chart · a pen, not a pencil · within the first seven days

- 01 ☐ **Colour every week you have lived.** One row per year of your life, fifty-two boxes per row, from your birth to this week. Do it by hand; the hour it takes is part of the instrument.
 - 02 ☐ **Draw a line at eighty.** Count the rows between this week and the line. Write the number of weeks on the ledger.
 - 03 ☐ **Run the tail-end sum for three people.** Take the three people your eulogy would name. For each: how many times a year do you see them, and for how many more years? Multiply. Write the three numbers on the ledger, next to their names.
 - 04 ☐ **Choose the weekly non-negotiable.** Two hours or more, same day, same time, for the work your governing lines require. Write it on the chart's legend, on the ledger, and in the calendar for the next twelve weeks.
 - 05 ☐ **Choose one weekly recurrence to stop.** Cancel it today. Write what it was and what it cost you per year.
-

WHY IT WORKS Countable time gets allocated; uncounted time gets spent. The non-negotiable turns a governing line into a recurring appointment, which is the only form in which a value survives contact with a calendar.

WHERE IT FAILS Filling the chart and making no decision. The chart is finished only when the calendar has twelve appointments in it and one recurring thing has been cancelled.

LEDGER · INSTRUMENT TWO

The allocation

WEEKS LIVED _____ WEEKS TO EIGHTY _____

PERSON _____ TIMES A YEAR YEARS LEFT TIMES MORE _____

My weekly non-negotiable: what, which day, which hours, where.

Twelve weeks, kept.

1	2	3	4	5	6	7
8	9	10	11	12		

What I stopped, and what it was costing me a year.

SIGNED _____ REVIEWED ON _____ KEPT / REVISED _____

INSTRUMENT THREE · THINKING

III

The Question

The quality of a life follows the quality of its questions.

Be patient toward all that is unsolved in your heart and try to love the questions themselves. Live the questions now.

RAINER MARIA RILKE, LETTERS TO A YOUNG POET, 1903

MONTH THREE

IN 1986 THE MATHEMATICIAN Richard Hamming gave a talk at Bellcore called *You and Your Research*, and the part people remember is a pair of questions he used to put to colleagues at lunch. What are the important problems in your field? And why aren't you working on them? He was not, he reported, welcome at that table afterwards. The questions were rude only because they were good. A good question is a search image: once you have it, you cannot stop seeing what it points at, and you cannot un-see what it makes obvious.

We are now surrounded by machines that will answer almost any question in seconds. This changes something fundamental. For most of history, the answer was the expensive part: it took a library, an expert, an experiment. Now the answer is nearly free, and the question is the only scarce input left. A person who asks better questions than the people around them has an advantage that compounds, because the answers arrive at the same speed for everyone.

This instrument is a way of generating questions on purpose, and of finding, among them, the one that matters.

— WHY IT WORKS

Questions direct attention, and attention, as the seventh instrument will argue, is the substance of experience. But there is a more specific finding. In the 1960s Jacob Getzels and Mihaly Csikszentmihalyi watched art students in a studio, given a table of objects and asked to produce a drawing. Some students picked up two objects and began. Others handled far more of them, rearranged them, changed their minds, and started late. The second group produced work judged more original, and when the researchers followed the students for years afterwards, it was the problem-finders, not the fast starters, who had become the successful artists. The quality of the drawing had been decided before the pencil touched the paper, in the choice of what to draw.

Most of us are fast starters. We take the first framing of a problem, usually the one somebody else handed us, and work hard inside it. The question *how do I get promoted* produces a different decade from *what work would I do if*

nobody could see me do it, and both are answerable. The framing is where the leverage is, and framing is a question.

Answers are now cheap. The question is the only scarce part.

There is also a structural reason to hunt for questions in bulk. The first ones you write are the ones you have already been asked. It takes twenty or thirty to exhaust the obvious, and it is only after that, when the easy material is gone, that a question appears that you have never let yourself ask. Every person who has done the exercise on the next page reports the same shape: the first twenty are dull, the next twenty hurt, and somewhere in the last ten is the reason they bought the book.

— HOW TO HOLD IT

Among the questions, look for one with a particular property: its answer settles the others. Engineers borrow a word from mathematics and call it the eigenquestion. *Should I take the job* may be settled by *do I want to be the kind of person that job produces*, which is settled by the eulogy. When you find the eigenquestion you will know, because the list gets shorter as you look at it.

A few forms are reliably useful. *What would have to be true for this to be the right choice?* turns an argument into a list of checkable facts. *What is the smallest experiment that would settle this?* turns a worry into a week. *Whose approval am I optimising for?* is usually answered by a name, and the name is usually a surprise. *What do I already know that I am pretending not to know?* is the one that hurts, and it is the one to carry. And for anything you have been calling complicated: *if I had to decide by Friday, what would I decide?* Most complicated things are simple things we would rather not say aloud.

Two older forms deserve a place. The engineers at Toyota asked *why* five times in a row, each answer becoming the next question, until a broken machine had been traced to a filter nobody had thought to clean. It works on lives as well as on factories: ask why you are tired five times and you rarely end up at sleep. And Paul Graham, writing about how great work gets done,

reduced the search for a vocation to one question: what are you excessively curious about, curious to a degree that would bore most other people? He added that curiosity never lies. Neither of these is a trick. They are questions shaped so that the honest answer is the only one that fits.

Leonardo da Vinci's notebooks are full of lists of questions, and they are the most human thing in them. *Describe the tongue of the woodpecker. Ask Benedetto Portinari how they walk on ice in Flanders. Ask the master of arithmetic to show you how to square a triangle.* He did not have answers to most of them, and he wrote them down anyway, because a question written down goes on working when the mind that wrote it has moved on. Keep yours in the commonplace book that the ninth instrument will ask you to start, and read the list once a month. Some of the questions will have answered themselves. Others will have become the work.

Do not answer the questions in the sitting. The exercise is to generate them and to sit with them, in Rilke's sense, for a month. Write the one you choose on the card and carry it. Questions carried for a month tend to answer themselves, quietly, usually while you are doing something else.

PRACTICE · INSTRUMENT THREE

Fifty questions

Sixty minutes · one sitting · a second sitting the next day · within the first seven days

- 01 ☐ **Write fifty questions about your own life.** Numbered, one sitting, no editing and no answering. Anything is admissible: work, money, the body, the people, the things you avoid. Do not stop before fifty. The first twenty are for warming up.
 - 02 ☐ **The next day, read them cold.** Circle the three that made you uncomfortable. Discomfort is the signal; it means the question is live.
 - 03 ☐ **For each of the three, write the answer you are pretending not to know.** One sentence each. If the sentence is easy, you circled the wrong question.
 - 04 ☐ **Find the eigenquestion.** Among the fifty, which single question, if answered, would settle most of the others? Write it on the ledger and on the card.
 - 05 ☐ **Carry it for the month.** Do not force an answer. Note on the ledger, on the last day, what it turned into.
-

WHY IT WORKS Problem-finding, not problem-solving, is what separated the artists who lasted from the ones who did not. Bulk generation exhausts the borrowed questions and exposes your own. A carried question recruits attention for a month at no cost.

WHERE IT FAILS Stopping at thirty. Answering as you go. Choosing a comfortable eigenquestion. If none of the fifty frightened you, write fifty more.

LEDGER · INSTRUMENT THREE

The questions

FIFTY WRITTEN ON _____

The three that made me uncomfortable, and the answer I am pretending not to know.

The eigenquestion.

What would have to be true.

On the last day of the month: what it turned into.

SIGNED_____
REVIEWED ON_____
KEPT / REVISED

INSTRUMENT FOUR · JUDGMENT

IV

Inversion

*Avoid stupidity before you attempt
brilliance.*

Man muss immer umkehren. Invert, always invert.

ATTRIBUTED TO CARL GUSTAV JACOB JACOBI, 1804–1851

MONTH FOUR

IN 1986 CHARLIE MUNGER was asked to give the commencement speech at the Harvard School in Los Angeles, and instead of telling the graduates how to succeed he told them, in detail, how to guarantee a life of misery. Be unreliable. Learn only from your own experience and never from anyone else's. Give up at the first setback. Think in a fog. He had borrowed the method from the nineteenth-century mathematician Jacobi, who solved hard problems by restating them backwards, and the speech is remembered when a thousand cheerful ones are forgotten because the backwards version is more useful. Everyone in the room already knew how to be unreliable. They had simply never heard it described as a strategy.

Inversion rests on an asymmetry that is easy to state and strange to accept. It is much easier to say what would wreck a life than what would perfect one, and the wrecking list is far more actionable, because every item on it is something you can stop. Nobody knows what makes a marriage great. Everybody knows what ends one. Medicine and aviation became safe not by discovering brilliance but by hunting down, one by one, the ways things go wrong.

So before this book asks you to build anything, it asks you to look for the failure. Not because failure is likely. Because it is knowable.

— WHY IT WORKS

In 1989 three researchers ran an experiment in which people were asked to explain a future event. Half were asked why it *might* happen. The other half were told it *had* happened and asked why. The second group produced about thirty per cent more reasons, and more specific ones. The psychologist Gary Klein turned this into a method he called the premortem: before a project begins, the team is told that it is a year later and the project has failed completely, and everyone writes down why. It works for two reasons. The certain past is easier to explain than the uncertain future, and the exercise gives everyone permission to say the thing they were going to be blamed for noticing later.

We need the help, because unaided we are absurdly optimistic. When students were asked how long their theses would take, they said thirty-four days

on average, and forty-nine if everything went as badly as it possibly could. The actual figure was fifty-five. Fewer than a third finished by the date they had named. The error was not in the estimates of the work. It was in the failure to imagine the ways the work would be interrupted, which is exactly what inversion supplies.

It is easier to say what would wreck a life than what would perfect one, and far easier to act on.

Try it on a year rather than a project. How would you guarantee that the next twelve months were wasted? Keep the phone beside the bed. Say yes to everything so that nothing gets your full attention. Wait until you feel ready. Optimise for looking busy. Keep score against people who are not playing your game. Postpone the conversation, the doctor, the apology. Learn only from your own mistakes. Never write anything down. Most people, reading the list, recognise their current arrangements.

— HOW TO HOLD IT

Three tools come out of inversion, and this month you will build all three. The premortem is the first. The second is the kill criterion, a phrase from the poker player Annie Duke: a signal, chosen in advance, at which you will stop. *If by June I have not found ten paying customers, I close it.* The point of choosing it now is that later you will be the wrong person to choose; you will be invested, tired, and telling a story. The person you are this month, who has not yet spent the year, is the only one who can be trusted to set the tripwire.

The third is the not-to-do list, which is the inverted goal. Goals are tiring; they require sustained effort against resistance. A not-to-do list requires only refusal, and refusal is available even on bad days. Five items, chosen from the wasted-year list, printed on the card, are worth more than most plans, because a plan tells you what to do when you are at your best, and a not-to-do list tells you what to do when you are not.

The institutions that have made the most progress with this instrument are the ones where failure is expensive. In 1935 the most advanced bomber in

the world crashed on a test flight because a highly experienced crew forgot to release a gust lock, and the response was not more training. It was a card listing the things that had to be done before take-off, which every pilot in the world now reads aloud. Seventy years later, when hospitals in eight countries adopted a nineteen-item surgical checklist, complications fell by more than a third and deaths by nearly half. The list did not contain a single brilliant idea. It contained the known ways to fail, and the discipline of checking that none of them was happening today.

One caution. Inversion is a tool for avoiding ruin, not a temperament. People who only invert become careful and small. The next instrument is about the other half: once the downside is bounded, how to be bold. Do this one first, because you cannot take many bets if any one of them can end the game.

PRACTICE · INSTRUMENT FOUR

Run the premortem

Forty-five minutes · the most important thing you are doing this year · within the first seven days

- 01 ☐ **Name the thing.** The project, the move, the relationship, the year itself. Write it at the top of a page.
 - 02 ☐ **Declare the failure.** Under it, write: *It is* [the date one year from today]. *It has failed completely.* Believe it for the next ten minutes.
 - 03 ☐ **List the reasons.** Every cause you can think of, specific, in the past tense. The first five will be obvious. Do not stop before fifteen.
 - 04 ☐ **Circle the three most likely,** and for each write one action this month that removes or weakens the cause. Put the actions in the calendar.
 - 05 ☐ **Set the kill criterion.** *If by* [date] [signal] *has not happened, I stop.* Write it on the ledger. It cannot be revised by the person you will be in June.
 - 06 ☐ **Write the not-to-do list.** Five things that would guarantee the wasted year, which you will not do. Copy them to the card.
-

WHY IT WORKS Prospective hindsight produces more and better reasons than forecasting. Kill criteria move the decision to quit to the one moment you can make it honestly. A not-to-do list works on days when nothing else does.

WHERE IT FAILS Listing causes you can blame on others. At least ten of the fifteen must be things you did or did not do.

LEDGER · INSTRUMENT FOUR

The premortem

THE THING _____

It failed because (the three most likely causes, and the action against each).

The kill criterion.

The not-to-do list.

SIGNED

REVIEWED ON

KEPT / REVISED



Asymmetry

Make many small bets whose downside is bounded and whose upside is not.

Most decisions aren't like that – they are changeable, reversible – they're two-way doors.

JEFF BEZOS, LETTER TO SHAREHOLDERS, 2016

IN HIS 2015 LETTER to shareholders, Jeff Bezos divided decisions into two kinds. Some are consequential and irreversible: one-way doors, to be taken slowly, with deliberation and consultation, because if you walk through and do not like what you see, you cannot get back. But most decisions are not like that. They are changeable, reversible, two-way doors, and they should be made quickly, by individuals or small groups, because the cost of a wrong two-way decision is small and the cost of slowness is not. His complaint was that large organisations use the heavy process on everything. So do people. We agonise over two-way doors as if they were one-way, and then, exhausted, walk through the one-way doors without looking.

Reversibility is the first sort, and it is quick: can I undo this in a month at a price I would pay? If yes, decide today. The second sort is the one this chapter is about, and it decides a life. For every choice, ask two questions. What is the worst that can happen, and is it bounded? What is the best, and is it?

Sending the letter, publishing the essay, asking the stranger, starting the small thing on Saturday mornings: the worst case is an hour and some embarrassment, and the best case has no ceiling. The shortcut, the lie, the uninsured risk, the borrowed money: the best case is a little more of what you already have, and the worst case is the end of the game. These are not the same kind of choice, and almost nobody treats them differently.

— WHY IT WORKS

The outcomes that shape a life are not evenly distributed. A career is made by one or two opportunities, a body of work by a handful of pieces, a life by a few people. The psychologist Dean Simonton found that across creative careers the ratio of hits to attempts is roughly constant: the most prolific creators produce the most masterpieces because they produce the most of everything, including failures. If you cannot know in advance which attempt will matter, the only strategy is many attempts, each cheap enough to survive.

But this only works if no single attempt can remove you from the game. Ruin is not a large loss; it is the loss after which there are no more bets. So the order matters, and it is the order of this book: bound the downside first, with the inversion you did last month, and then be as bold as the bound allows.

The people who seem reckless and keep winning are usually not reckless. They have simply made sure that nothing they try can kill them, and then tried everything.

	UPSIDE BOUNDED	UPSIDE UNBOUNDED
Downside bounded	Chores. Two-way doors. Decide today, or delegate.	Bets. Publish, ask, ship, show. Take as many as you can afford.
Downside unbounded	Ruin. The shortcut, the lie, the uninsured. Never.	Wagers. The company, the move. Size small; never what you cannot lose.

Avoid ruin. Then take many small bets. Everything else is detail.

— HOW TO HOLD IT

The enemy of this instrument is a habit the poker player Annie Duke calls resulting: judging a decision by its outcome. A good bet that lost was still a good bet; a bad bet that won is still a bad bet, and the win will teach you to make it again. The only defence is to record the decision at the time it is made, before the outcome exists. Peter Drucker recommended exactly this: whenever you make a key decision, write down what you expect to happen, and nine or twelve months later compare. He said he had done it for twenty years and that it was the only reliable way he knew to find out what he was actually good at. Benjamin Franklin, in 1772, described dividing a sheet into two columns, pro and con, adding to it over several days, and striking out reasons of equal weight until the balance showed. He called it moral algebra. It is the oldest decision instrument in this book, and the ledger opposite is its descendant.

For the rare one-way door, borrow a question from the third instrument: what would have to be true for this to be right? Then find the cheapest way to check each thing before you walk through. Most one-way doors have a two-way door standing next to them: a trial, a sabbatical, a smaller version, a conversation. Take that one first.

The mathematics of this was worked out at Bell Labs in 1956 by John Kelly, who showed that a bettor with an edge should stake a fixed fraction of their bankroll, never the whole of it, and that the fraction which grows wealth fastest is also the one that makes ruin impossible. Ed Thorp used the formula at blackjack tables and then in the markets for thirty years. Venture investors use a cruder version of the same idea: a fund makes twenty bets expecting most to fail, because one or two will return more than all the rest together, and the only unforgivable error is to be unable to make the twenty-first. A life is a fund of this kind. The people who look lucky are the ones who were still solvent when their bet came in.

And size the bets. You do not need the mathematics of the Kelly criterion to keep its spirit: never stake the amount, of money or time or reputation, that would make the next bet impossible. Small, many, survivable, and let the arithmetic of the tail do the rest.

PRACTICE · INSTRUMENT FIVE

Sort the bets

Sixty minutes · then ten minutes each week · within the first seven days

- 01 ☐ **List every decision you have been putting off.** Ten minutes, no filter. Include the small ones; they are the ones that have been costing you.
 - 02 ☐ **Mark each door.** One-way or two-way: can it be undone in a month at a price you would pay? Decide every two-way door today, and write the date you will review it.
 - 03 ☐ **For each one-way door, write what would have to be true for it to be right, and the cheapest way to find out.** Put the finding-out in the calendar.
 - 04 ☐ **List ten small bets.** Bounded downside, unbounded upside: the letter, the essay, the ask, the small thing on Saturday. Place three this month.
 - 05 ☐ **Open the decision ledger.** For every significant decision this month, write what you expect to happen and a date to check. Review the entries in nine months. This is the one habit on the page that should outlive the book.
-

WHY IT WORKS Reversible decisions made fast free the attention that irreversible ones need. Hits scale with attempts when no attempt can end the game. A written expectation is the only cure for resulting.

WHERE IT FAILS Treating a wager as a bet. If the downside is your savings, your health or your name, it is not small, however good the story.

LEDGER · INSTRUMENT FIVE

The decision ledger

OPENED ON _____

Two-way doors, decided today.

The one-way door, and what would have to be true.

Three bets placed this month.

DECISION	WHAT I EXPECT TO HAPPEN	REVIEW ON

SIGNED

REVIEWED ON

KEPT / REVISED

VI

The Loop

You become whatever you get fast feedback on.

The first principle is that you must not fool yourself – and you are the easiest person to fool.

RICHARD FEYNMAN, CALTECH, 1974

A TWELVE-YEAR-OLD will learn a difficult video game in a week and fail to learn long division in a year, and the difference is not motivation. In the game, every action is followed within a second by an honest signal about whether it worked. In the classroom, the signal arrives in a fortnight, in red ink, about something the child no longer remembers doing. Call the time between an action and an honest signal about it the loop length. It predicts how fast a person learns better than anything else we can measure, including the thing we call talent.

Most adult skills have loops measured in months or years. You give the presentation and never learn what the room thought. You make the hire and find out in a year. You write in the way you have always written because nobody has ever told you, specifically and soon, where a reader got lost. In such conditions people do not improve with experience; they simply repeat their first year many times and become confident.

This instrument is a way of taking the loops that matter and making them short.

— WHY IT WORKS

The research on expertise is usually summarised as ten thousand hours, which is a misreading of a 1993 study of violinists that its own author spent the rest of his life correcting. What the study actually found was that the best players had spent more time in a particular kind of practice: designed, effortful, aimed at a specific weakness, with immediate feedback. Hours alone predicted little. A later meta-analysis found that deliberate practice explained about a quarter of the variation in performance in games and a fifth in music, but only four per cent in education and less than one per cent in the professions. That last figure is often quoted as evidence that practice does not matter at work. Read it the other way. Games and music have loops of seconds. The professions have loops of years, so long that practice cannot find anything to grip.

The best-studied loop is the smallest. Students who learned a list of foreign words and then repeatedly tested themselves recalled about eighty per cent a week later. Students who spent the same time re-reading the list rec-

alled about a third. Re-reading feels like learning and is not; the test, which feels like failure, is the learning. Spacing the tests out over days beats cramming them, by a wide margin, in every study that has looked. And the forecasters in Philip Tetlock's tournaments who beat professional analysts did it mostly by keeping score, updating often, and in small steps. They had closed a loop that their rivals had left open.

| Talent is a rumour. Loop length is a measurement.

— HOW TO HOLD IT

A loop has three parts, and each can be shortened. The action can be made smaller: a paragraph reviewed daily teaches more than a chapter reviewed monthly, because it is reviewed thirty times. The signal can be made sooner: show the draft, the prototype, the plan, before it is finished, to someone whose job is not to like you. And the signal can be made honest: a count of readers is a signal, a count of likes is not; a friend's praise is not, a stranger's confusion is. Feedback that is fast but flattering is worse than none, because it trains you confidently in the wrong direction.

Two habits follow. The first is to test yourself instead of re-reading, which applies to this book: on the last day of each month, before opening the chapter, write from memory what the instrument said. Then check. What you could not retrieve, you had not learned. The second is to keep score in writing, which is why every ledger in this book ends with a signature and a date. A record you have to sign is a loop closed.

Two cautions. The first is that a measure, once it becomes a target, stops measuring: the economist Charles Goodhart noticed this in banking and it holds everywhere, from word counts to step counts. When the number starts to feel like the point, change the number. The second is that the fastest honest loop is almost always a person. A coach, an editor, a customer, a child who does not understand your explanation. Software can tell you what happened; only a person can tell you what it was like, and it is usually what it was like that you needed to know.

The surgeon Atul Gawande, eight years into practice and with his results plateaued, did something surgeons do not do: he asked a retired professor to stand in his operating theatre and watch. The first session produced a page of notes on things Gawande had not known he was doing, from where he stood to how he held the light, and over the following months his complication rate, which had stopped improving, began to fall again. Every performer with a coach has a loop measured in minutes. Almost everyone else, once the formal training ends, has a loop measured in years, and calls the resulting silence experience.

The instrument this month is arithmetic. Find the longest loop attached to something your governing lines require, and cut it by ten. If you learn how the meeting went once a quarter, learn it weekly. If you find out whether the writing works once a year, find out on Friday. The rate at which you become the person in the eulogy is set by the length of these loops, and almost all of them are longer than they need to be.

PRACTICE · INSTRUMENT SIX

Shorten the loop

Thirty minutes to design · thirty days to run · begin within the first seven days

- 01 ☐ **Name the three skills your governing lines require.** Not roles: skills. Writing clearly, running a room, staying calm with a child, selling, listening.
 - 02 ☐ **Measure the loop.** For each, write how long currently passes between doing it and knowing honestly how it went. Days, weeks, never.
 - 03 ☐ **Take the longest and design a loop ten times shorter.** Who gives the signal, in what form, how often. A person, a test, a number, a recording of yourself. Write the design on the ledger.
 - 04 ☐ **Run it for thirty days.** Mark each iteration on the grid. Aim for thirty. Change the design on day ten if it is not producing honest signal.
 - 05 ☐ **Retrieve, do not re-read.** On the last day of every month from now on, write what the chapter said before you open it. Then open it and grade yourself.
-

WHY IT WORKS Testing beats re-reading by a factor of two in retention; spacing beats massing; fast honest signal is what deliberate practice is made of. Shortening a loop by ten multiplies the iterations in a year by ten.

WHERE IT FAILS Choosing a signal that is fast and kind. If the loop never tells you anything you did not want to hear, it is not a loop. It is applause.

LEDGER · INSTRUMENT SIX

The loop

DESIGNED ON _____

SKILL	LOOP NOW	LOOP DESIGNED: WHO, WHAT FORM, HOW OFTEN

Thirty iterations.

What the signal said that I did not want to hear.

From memory, on the last day: what this instrument says.

_____ SIGNED	_____ REVIEWED ON	_____ KEPT / REVISED
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VII

Attention

Your experience is what you agree to attend to.

My experience is what I agree to attend to. Only those items which I notice shape my mind.

WILLIAM JAMES, THE PRINCIPLES OF PSYCHOLOGY, 1890

A WAKING DAY is about a thousand minutes. Sixteen and a half hours, a little more or less depending on how the eleventh instrument is going, and then it is over and another begins. Ask most people where yesterday's thousand went and they can account, with effort, for about seven hundred. The rest were spent, in the strict sense of the word, and nobody knows on what. That missing three hundred is three hundred a day, two thousand a week, a hundred thousand a year: two and a half months of waking life, every year, unaccounted for.

William James wrote that his experience was what he agreed to attend to, and he meant it literally. Attention is not a productivity technique. It is the material that a life is made of, because whatever you did not attend to, you did not experience, whoever you were with and wherever you were. This is why the phone matters. It is not that it wastes time. It is that it purchases, at a price of zero, the only thing you have, and it does so with the best engineering of the age. Herbert Simon saw this coming in 1971: a wealth of information, he wrote, creates a poverty of attention.

The instrument is a grid of a hundred boxes, ten minutes each, and it will tell you where the thousand go.

— WHY IT WORKS

The cost of divided attention is larger than it feels. In a series of experiments, Sophie Leroy showed that when people switch from one task to another, part of their attention stays behind on the first: she called it attention residue, and it measurably degrades performance on the second task. The residue is worst when the first task was left unfinished, and it shrinks when the person had reached a point of closure before switching. Every glance at a message is a switch, and every switch leaves residue, so a day of small interruptions is not a day with some minutes missing. It is a day in which every remaining minute was done worse.

The device does not even need to be used. Researchers who asked people to complete tests of working memory found that performance was lowest when their own phone was on the desk, higher when it was in a pocket or bag, and highest when it was in another room, even though in every condit-

ion it was switched off. The brain spends capacity on not looking at it. Willpower, in other words, is the wrong tool; distance is the right one. Environment beats intention, every day, for everyone.

Where the thousand minutes go is not a productivity question. It is the whole question.

One more finding cuts the other way. People given a boring, undemanding task during a break solved more creative problems afterwards than people who rested or worked hard. An attentive life is not a life of unbroken concentration; it needs empty time, walks without input, the shower and the queue. The phone has abolished those too. It is not only your focus it takes. It takes your boredom, which is where the ideas were.

— HOW TO HOLD IT

Begin by counting. For three days, each evening, reconstruct the day in ten-minute boxes and put one word in each. Then colour the boxes you would choose again. The exercise is not to feel guilty; it is to see. Most people find that the unchosen boxes cluster in the same two or three places, and that they were not enjoying them.

Then choose the sacred hours: two hours a day, at the same time, for the work your governing lines require, with the phone in another room and one task on the desk. Not face down. Another room. Cal Newport calls this deep work, professional effort in a state of distraction-free concentration that pushes your abilities to their limit, and observes that it is rare, valuable and getting rarer. Two hours a day of it, for a year, is more than most people manage in a career. End each session by writing where you stopped and what comes next, because a task left with a known next step leaves less residue than one abandoned mid-thought.

Finally, remove one input for the month. One app, one feed, one source of news. Delete it, and on the last day, if you miss it, reinstall it. You will not. The boxes it occupied will fill with something, and the exercise is to notice what.

The scale of the problem is measurable. When a research firm installed a tool on volunteers' phones to count every tap, swipe and click, the average user touched the device more than two and a half thousand times a day, and the heaviest users more than five thousand. Nobody chose that number. It is what a device optimised for touching produces when it is within reach, and it is why the instruction is a room, not a resolution.

One last observation, which belongs as much to the twelfth instrument as to this one. The violinists in the famous practice study could not manage more than about four hours of real concentration a day, and the best of them protected those hours and napped in the afternoon. Attention is finite, and what is left of it after the work is the part you give to other people. Simone Weil wrote that attention is the rarest and purest form of generosity. Most of us give the sacred hours to a screen and the residue to the people at dinner. The grid will show you whether that is true of you, and it is the one thing on it that is worth changing first.

PRACTICE · INSTRUMENT SEVEN

The thousand minutes

Ten minutes each evening for three days · then two hours a day · begin within the first seven days

- 01 ☐ **Reconstruct three days in ten-minute boxes.** Each evening, fill a hundred boxes with one word each: work, phone, child, walk, meeting, nothing. Use the grid on the ledger for the third day; plain paper for the first two.
 - 02 ☐ **Colour the boxes you would choose again.** Count the ones you cannot account for. Write the number.
 - 03 ☐ **Set the sacred hours.** Two hours, same time daily, one task, phone in another room. Write the hours on the ledger and defend them like a flight.
 - 04 ☐ **Close every session.** Before you stop, write one line: where I stopped, and the next step. This is what stops the residue following you into the evening.
 - 05 ☐ **Remove one input for the month.** Delete it today. Write what it was. On the last day, write what filled the space.
-

WHY IT WORKS Counted minutes get chosen; uncounted minutes get taken. Distance from the device restores capacity that intention cannot. Closure removes residue. Empty time restores the incubation that ideas need.

WHERE IT FAILS Sacred hours that move. If the hours change daily, they are not sacred; they are available, and something will take them.

LEDGER · INSTRUMENT SEVEN

The thousand minutes

DAY THREE, THE _____

One hundred boxes of ten minutes, from waking to sleep.

BOXES I COULD NOT ACCOUNT FOR _____ SACRED HOURS, FROM
_____ TO _____

The input I removed, and what filled the space.

_____ SIGNED	_____ REVIEWED ON	_____ KEPT / REVISED
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INSTRUMENT EIGHT · PATIENCE

VIII

Compounding

*Small, repeated, for years, beats large,
occasional, for months.*

Play long-term games with long-term people.

NAVAL RAVIKANT, 2018

MONTH EIGHT

THE ARITHMETIC OF COMPOUNDING is usually presented as a motivational trick: get one per cent better every day and you will be thirty-seven times better in a year. The sum is correct and the premise is false, because nobody improves one per cent a day at anything for a year. The honest version is slower and more interesting. One per cent a week, which is achievable, makes you two thirds better in a year, which is unimpressive. It makes you thirteen times better in five years, and a hundred and seventy-six times better in ten. The curve is flat for a long time and then it is vertical, and the whole difficulty of compounding is that people leave during the flat part, because from inside the flat part it looks like nothing is happening.

WEEKLY GAIN	ONE YEAR	FIVE YEARS	TEN YEARS	TWENTY YEARS
0.5 %	1.3×	3.7×	13×	179×
1 %	1.7×	13×	176×	31,000×
2 %	2.8×	172×	29,700×	–

Not everything compounds. Attention spent on the news does not; neither do credentials nobody checks, or work that leaves nothing behind when it is done. What compounds is anything that makes the next unit easier or more valuable than the last: a skill, a body of work, a reputation, a set of relationships built on kept promises, a body that is looked after, money that is left alone. And what compounds against you compounds just as reliably: debt, resentment, the unread messages, the back.

— WHY IT WORKS

The mechanism is mathematical and the obstacle is human. We discount the future steeply, and we discount our future selves most of all, treating the person who will be alive in ten years as a stranger whose problems are their own. The experiment mentioned in the first chapter bears repeating: shown an aged image of their own face, people more than doubled what they set aside for that person. The future self has to be made vivid before we will invest in it, and the way to make it vivid is to let it speak.

Two other ideas belong here. Cal Newport calls the stock of rare and valuable skills a person has built their career capital, and argues that the things everyone wants from work, autonomy and meaning and control over one's time, are bought with it and with nothing else. And Naval Ravikant pointed out in 2018 that a single person now has access to forms of leverage that once needed an organisation: capital, other people's effort, and, newest of all, code and media, products that cost nothing to copy and work while you sleep. What he called permissionless leverage has, since he wrote, grown by an order of magnitude. A person with a skill, a machine, and ten years is a different kind of thing than they were in any previous century.

The curve is flat for years and then it is vertical. Most people leave during the flat part.

The practical consequence is that the important question about any use of a week is not whether it is productive but whether it compounds. An hour that leaves nothing behind is an hour. An hour that adds to something is an hour plus a fraction of every hour that follows.

— HOW TO HOLD IT

Choose one compounding asset. Not five: one. A skill, a body of work, a small set of relationships, a financial habit. The weekly non-negotiable you set in the second month is its vehicle, and the sacred hours of the seventh are its fuel. Then measure inputs, not outcomes, for as long as the curve is flat: weeks kept, not results achieved, because the results will not appear on any schedule you can plan for, and if you wait for them you will quit in year two, which is when almost everyone does.

The instrument this month is a letter. It is written by you, dated ten years from today, to the person you are now, and it describes what compounded: what the asset became, what it required every week, what it cost, what it made possible. Written honestly, the letter does two things. It makes the future self vivid enough to invest in. And it usually reveals, in the writi-

ng, that the weekly requirement is smaller than you feared, and the only hard part is not stopping.

Two illustrations, one famous and one useful. Warren Buffett has been investing since he was eleven and is one of the richest people who ever lived, and well over ninety-nine per cent of his fortune arrived after his fiftieth birthday. He was not much better at fifty than at forty. The curve had simply reached the vertical part. The useful illustration is a rule of thumb known as the Lindy effect: a thing that has lasted a long time can be expected to last about as long again. Choose assets accordingly. Clear writing, selling, listening, arithmetic and the ability to sit alone with a hard problem have compounded for people for centuries and will compound for you. The tool that was invented last year may or may not.

Charlie Munger's first rule of compounding was never to interrupt it unnecessarily. The book cannot tell you what your asset is. It can only insist that you name one, protect it, and let the years do the arithmetic.

PRACTICE · INSTRUMENT EIGHT

The ten-year letter

Sixty minutes · by hand · kept inside this book · within the first seven days

- 01 ☐ **Name the asset.** One skill, body of work, set of relationships or financial habit which, held for ten years, changes everything else. Write it on the ledger in one line.
 - 02 ☐ **Write the letter.** Dated ten years from today, from you to the person reading this now. One page. What compounded, what it demanded each week, what you gave up, what it made possible. Do not flatter yourself; the letter is more useful if it is plausible.
 - 03 ☐ **Set the weekly minimum.** The smallest unit you could sustain for five hundred and twenty weeks. Three hours. Five hundred words. One call. Write it and put it inside the non-negotiable from month two.
 - 04 ☐ **Name what compounds against you,** and start reducing it this month. Write the first step.
 - 05 ☐ **Fold the letter and keep it at this page.** The sealed envelope will ask about it.
-

WHY IT WORKS A vivid future self more than doubles what people invest in it. Measuring inputs keeps you in the game through the flat years. A named asset with a weekly minimum is a compounding process; a vague ambition is not.

WHERE IT FAILS Choosing three assets. Anything that competes for the same weekly hours halves the rate. One.

LEDGER · INSTRUMENT EIGHT

The asset

LETTER DATED _____

The asset, in one line.

The weekly minimum, and where in the week it lives.

The first lines of the letter.

What compounds against me, and the first step against it.

SIGNED

REVIEWED ON

KEPT / REVISED

INSTRUMENT NINE · CREATIVITY

IX

Taste

Quantity is the road to quality.

To create consists precisely in not making useless combinations and in making those which are useful and which are only a small minority. Invention is discernment, choice.

HENRI POINCARÉ, SCIENCE AND METHOD, 1908

MONTH NINE

THERE IS A STORY in a small book called *Art & Fear* about a ceramics teacher who divided a class in two. One half would be graded on quantity: fifty pounds of pots for an A, forty for a B. The other half would be graded on quality, and needed to produce only one pot, but a perfect one. At the end of the term, the best pots all came from the quantity group. They had made hundreds, learned from each, and got better. The quality group had sat theorising about perfection and produced, in the end, a great deal of theory and some mediocre clay. The story is a parable; one of the authors later admitted it was borrowed from a photography class and the medium changed. But the finding it dramatises is real and has been measured.

Dean Simonton, who has spent a career studying creative output, found what he called the equal-odds rule: across a career, the ratio of a creator's hits to their total output stays roughly constant. Quality does not rise as a fraction of quantity. The people who produced the most masterpieces were the people who produced the most work, and they also produced the most failures, because the failures and the masterpieces come from the same process. There is no known way to make only the good ones.

This is the instrument for anyone who has ever wanted to make something and found that what they made was not as good as what they could see. That gap has a name.

— WHY IT WORKS

Ira Glass gave it the name. Everyone who does creative work, he said, gets into it because they have good taste. But for the first years the work is not as good as the taste, and the taste, which is what brought you in, is exactly what makes the work disappointing. Most people quit in that gap. The only way through it is a volume of work. Taste, in other words, is what you have when your standards outrun your hands, and the cure is not better standards.

Poincaré described creation as combination followed by choice. Invention, he said, is discernment among combinations, most of which are useless. Both halves need volume. Combination needs raw material, which is why every creative tradition has kept a commonplace book: John Locke published a method for one in 1706, Leonardo filled thousands of pages, and the

emperor Marcus Aurelius wrote the notes to himself that we call the *Meditations* with no intention of anyone reading them. And choice needs candidates. You cannot discern among three ideas. You can among a hundred.

Taste is what you have when your standards outrun your hands. The only known cure is volume.

There is one more reason quantity works, and it connects to the sixth instrument. Each piece made and shown is a loop closed. The person who makes fifty small things in a year has had fifty honest signals. The person who spent the year on one large thing has had none, and has also, somewhere around month eight, started to protect it from the signal it needs.

The history of the people we call geniuses is mostly a history of volume. Edison was granted more than a thousand patents, most for things nobody remembers. Bach wrote well over a thousand works, many of them for the following Sunday. Picasso left tens of thousands of pieces. We remember the hits and forget that the hits were a small, fixed fraction of a very large number, and that the people who made them had no better idea than you do which ones would last.

— HOW TO HOLD IT

Separate generation from judgment. Never edit in the same sitting as you produce; the critic and the maker are different people and they should not be in the room together. Generate in bulk, on a timer, and judge tomorrow. Ship small things on a schedule, weekly if you can, and show each one to somebody outside your house. Keep the commonplace book: one page a day, in your own hand, of sentences, ideas and observations that struck you, because the combinations that will matter in year three are made of what you collected in year one.

Use constraints on purpose. A form, a word count, a deadline, a single colour, a fixed hour. The blank page is not freedom; it is the absence of a problem, and problem-finding, as the third instrument said, is where the

le it is twenty to one, and no amount of taste survives it. This month the ratio is one to one: an hour made for every hour consumed.

Montaigne had the beams of his library painted with sentences from the books he loved, so that he could not look up from his desk without meeting one, and the essays he wrote in that room, which invented a form, are made largely of those sentences argued with. That is what the commonplace book is for. It is not an archive. Read it once a month and the entries begin to combine, and the combinations are the ideas that you will later believe you had.

And do not polish the first version. Make the second. Almost everything that is good about a piece of work arrives in the rewrite, when the maker can finally see what the thing wanted to be, and almost everything wrong with a first draft is fixed faster by a second draft than by any amount of attention to the first. This is quantity again, at the scale of a single piece.

Finally, do not wait to be good. The gap does not close by waiting; it closes, in Glass's words, by going through a volume of work, and there is no way to get to the other side except through the middle.

PRACTICE · INSTRUMENT NINE

One hundred ideas

Ninety minutes in one sitting · then one small thing a week · begin within the first seven days

- 01 ☐ **Write one hundred ideas.** For the work your governing lines require: projects, essays, products, experiments, gifts, questions, changes. Numbered, one sitting, no judging. The first thirty are everybody's ideas; the last thirty are yours.
 - 02 ☐ **The next day, circle five.** Choose by asymmetry: cheap to try, no ceiling if it works. Write them on the ledger.
 - 03 ☐ **Make the smallest version of one this week,** and show it to one person outside your house. Then one a week for the rest of the month.
 - 04 ☐ **Start the commonplace book.** Any notebook. One page a day: sentences copied by hand, observations, connections. Date every page.
 - 05 ☐ **Hold the ratio at one to one.** An hour made for every hour consumed. Keep a rough tally on the ledger.
-

WHY IT WORKS Hits scale with attempts; discernment needs candidates; each small thing shown is a loop closed. The commonplace book supplies the raw material that combination runs on.

WHERE IT FAILS Editing during generation, and making the first idea instead of the hundredth. If the hundred took forty minutes, you stopped at the obvious ones. Do it again.

LEDGER • INSTRUMENT NINE

The hundred

WRITTEN ON _____

The five circled.

WEEK	THE SMALL THING MADE	SHOWN TO
1		
2		
3		
4		

Three sentences copied into the commonplace book this month.

HOURS MADE _____ HOURS CONSUMED _____

SIGNED _____

REVIEWED ON _____

KEPT / REVISED _____

X

The Ask

Most of the world is a door that opens if you knock.

Ask, and it shall be given you; seek, and ye shall find; knock, and it shall be opened unto you.

MATTHEW 7:7

THE PSYCHOLOGIST RICHARD WISEMAN once gave a newspaper to people who described themselves as lucky and to people who described themselves as unlucky, and asked them to count the photographs in it. On the second page, in type two inches high, was a message: *Stop counting. There are 43 photographs in this newspaper.* The unlucky people took about two minutes. The lucky people took seconds. Further in was a second message, offering a cash prize to anyone who told the experimenter they had seen it. The unlucky group, heads down, counting, missed that too. It is a demonstration rather than a controlled study, but it illustrates what Wiseman found over a decade of interviews. Lucky people are not luckier. They notice more, they talk to more people, and they ask.

Asking is the cheapest bet in the fifth instrument's table. The downside is a moment of discomfort, bounded and brief; the upside is a job, a mentor, a customer, an apology accepted, a door you did not know existed. And yet most people ask for almost nothing, because they are certain, without ever having tested it, that the answer will be no. The evidence says they are wrong, by about half.

— WHY IT WORKS

In a series of experiments at Columbia and Stanford, Francis Flynn and Vanessa Bohns asked people to predict how many strangers they would have to approach to get a given number to agree to a small request: fill in a questionnaire, walk them to a building, lend a phone. The predictions were about twice the real figure. People systematically underestimate how likely others are to say yes, because they focus on the cost of helping and forget that the person asked is also weighing the discomfort of refusing. Refusal is awkward. Most people would rather help.

The second finding is older and stranger. In 1973 Mark Granovetter asked professionals in Boston how they had found their jobs, and among those who had found them through a person, more than four in five had found them through someone they saw occasionally or rarely: a former colleague, a friend of a friend, a name from years before. Close friends know what you know. It

is the weak ties, at the edges of your network, who know something you do not, and they will tell you only if you ask.

The answer to the question you do not ask is always no, and it is the only no that is guaranteed.

There is even a finding that asking improves the relationship. In a 1969 experiment, people who were asked to do a favour for the experimenter afterwards liked him more than those who were not asked. Benjamin Franklin had described the effect two centuries earlier, having borrowed a rare book from a political rival in order to make a friend of him. We like the people we have helped, because we have to explain to ourselves why we helped them.

— HOW TO HOLD IT

Ask well. A good ask is specific, small enough to say yes to in a minute, and comes with a way out: *no is completely fine*. It says who you are in one line, what you have done that is relevant in one line, and what you want in one line, and it stops. It does not apologise for existing. And it asks for the thing itself, not for permission to ask for it later, because *could I pick your brain sometime* is a request for a meeting about a request.

Asking upward has its own grammar. The people whose help would change things receive many requests and answer the ones that show work: I read the thing you wrote, I tried what it suggested, this is what happened, here is my one question. Effort is the currency, and a specific question that took you an hour to earn is answered more often than a general one that took a minute. And when the answer is no, the relationship is almost never damaged by it. People remember being asked well. What they remember badly is being asked vaguely, or not at all, by someone who later turned out to have needed them.

Then change what you count. In 2012 a man named Jia Jiang spent a hundred days making requests designed to be refused, to cure himself of the fear of rejection, and found that a surprising number were accepted and that the refusals, once collected deliberately, stopped hurting. Collect the no's. Set a

number for the month, ten, and go and get them. You cannot reach ten no's without collecting several yeses on the way, and by the tenth no the instrument has done its work, because the thing you were afraid of has happened repeatedly and you are fine.

When Steve Jobs was twelve he wanted to build a frequency counter and needed parts, so he looked up Bill Hewlett, the founder of Hewlett-Packard, in the telephone directory and called him at home. Hewlett talked to him for twenty minutes, gave him the parts, and gave him a summer job. Jobs told the story for the rest of his life, and the moral he drew was not about Hewlett. Most people, he said, never pick up the phone and call. Most people never ask. And that, sometimes, is what separates the people who do things from the people who dream about them.

One ask is different from the others, and it is on the practice page. There is a letter you have been not writing. Everyone has one: the thanks never given, the apology, the request to the person whose help would change things. Write it this month, by hand, and send it. It is the highest-asymmetry bet in the book.

PRACTICE · INSTRUMENT TEN

Five asks in seven days

Twenty minutes to list · seven days to ask · one letter · begin within the first seven days

- 01 ☐ **List ten asks you have been avoiding.** A person, a thing, a sentence. Introductions, rates, help, forgiveness, a seat at a table.
 - 02 ☐ **Rank them by asymmetry.** Bounded downside, unbounded upside first. Pick five.
 - 03 ☐ **Make the five within seven days.** Specific, short, with a way out. Write on the ledger what the answer was, not what you feared it would be.
 - 04 ☐ **Write the letter you have been not writing.** Five hundred words, by hand, to the person it is for. Send it or deliver it. Write on the ledger to whom, and when.
 - 05 ☐ **Collect ten no's before the month ends.** Tally them. Note what each one taught you about how to ask.
-

WHY IT WORKS People underestimate compliance by about half; information lives in weak ties; being asked makes people like you more, not less; deliberately collected refusals stop hurting. Every ask is a bet with bounded downside.

WHERE IT FAILS Asking vaguely, asking for permission to ask, and asking only people who cannot say no. If all five were easy, they were not the five.

LEDGER · INSTRUMENT TEN

The asks

LISTED ON _____

THE ASK	TO WHOM	THE ANSWER
---------	---------	------------

The letter: to whom, and the day it was sent.

Ten refusals, collected.

1	2	3	4	5	6	7
8	9	10				

What one yes opened.

SIGNED _____	REVIEWED ON _____	KEPT / REVISED _____
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INSTRUMENT ELEVEN · ENERGY

XI

The Body

The mind is something the body does.

Subjects were largely unaware of these increasing cognitive deficits.

VAN DONGEN, MAISLIN, MULLINGTON & DINGES, SLEEP, 2003

MONTH ELEVEN

IN 2003 A TEAM at the University of Pennsylvania put forty-eight healthy adults in a laboratory for two weeks and allowed them four, six or eight hours in bed a night. The people on six hours, which is roughly what a great many working adults consider normal, declined steadily for the whole fortnight. By the end, their attention and working memory had degraded as much as if they had gone without sleep entirely for up to two nights. That is not the finding. The finding is in the sentence on the opposite page: they rated themselves only slightly sleepy, and were largely unaware of the deficit. The instrument that judges whether you are impaired is the one that is impaired.

Every instrument in this book runs on a brain, and a brain is something a body does. It is fed, slept, moved and lit, or it is not, and when it is not, the person carrying it does not feel stupid. They feel normal, and they make normal-feeling decisions that a rested version of them would not make. You are not tired. You are under-slept, and the difference is that tiredness is felt and under-sleep is not.

This chapter is not about health, in the usual sense. It is about judgment, and about the four cheapest ways to improve it that have ever been found.

— WHY IT WORKS

Sleep is the first, and the research above is enough. The second is movement. In a one-year trial, older adults assigned to walk briskly three times a week grew the memory-forming part of the hippocampus by about two per cent, reversing one to two years of ordinary age-related shrinkage, while the control group, who stretched, lost more than one per cent. The review literature is titled without irony: be smart, exercise your heart. Aerobic movement is the closest thing we have to a drug for attention, and it is free.

The third is light. The body's clock is set by daylight in the morning, and a clock that is set runs everything else on time: alertness in the day, sleep at night, the hormones in between. Ten minutes outdoors within an hour of waking is the whole prescription, and most people, who now wake in dim rooms and look at bright screens at night, have it exactly backwards.

You cannot feel the deficit. That is the finding. The tired brain rates itself fine.

The fourth is breath, and here the evidence is newer. In 2023 a Stanford trial compared five minutes a day of several breathing patterns against five minutes of mindfulness meditation. Over a month, breathing produced the larger improvement in mood, and the most effective pattern was the simplest: a double inhale through the nose followed by one long exhale through the mouth, repeated slowly. It is the pattern the body already uses, spontaneously, before sleep and after crying. Five minutes of it lowers arousal reliably enough to be used before a decision, a conversation or a night's sleep.

— HOW TO HOLD IT

The four are not negotiable, which is the point: they are decided once and then not decided again. A fixed wake time, seven days a week, and an alarm for going to bed, not for getting up, set seven and a half hours before it. Daylight within an hour. Thirty minutes of movement a day, three of them hard enough to make conversation difficult. Five minutes of the breathing pattern when arousal is high or sleep is not coming. Written down, they take four lines. Kept for fourteen days, they change the quality of every other instrument in the book, because every other instrument is a form of judgment.

Two substances undo the first item more often than anything else. Caffeine has a half-life of around five to six hours, so a cup at four in the afternoon is half a cup at ten, still in the blood when the head is on the pillow. Alcohol in the evening brings sleep on faster and then fragments the second half of the night, which is where most of the useful sleep was. Neither has to be given up. Both have to be moved: caffeine to the morning, alcohol to the occasions rather than the evenings. And the seventh instrument applies here: the phone sleeps in another room, the bedroom is dark and cool, and the decision is made once, by arrangement, rather than nightly, by willpower.

Do not add to the list. There is an industry that would like to sell you a fifth, sixth and twentieth item, and most of them are at best harmless. The

four above are the ones with evidence behind them that any adult can act on for nothing. If you have a condition, a clinician outranks this chapter; the chapter's claim is modest and it is this. For most people, most of the time, these are the highest-return hours in the week, and the return arrives in the currency every other chapter spends.

The thirty minutes of movement need not be taken from the working day; they can be part of it. When Stanford researchers gave people tests of creative thinking sitting down and then walking, output rose by an average of sixty per cent on foot, and the effect persisted for a while after sitting back down. A walk is not a break from thinking. For a large class of problems it is the best available way of doing it, which is why so many of the people whose work lasted walked every day and said so.

Run it as a loop. Fourteen days, the grid on the ledger, and on the fifteenth day a written answer to a single question: what changed? The answer is usually not in the body. It is in the decisions.

PRACTICE · INSTRUMENT ELEVEN

Fourteen days of the four

Four lines to decide · fourteen days to keep · begin within the first seven days

- 01 ☐ **Fix the wake time.** Seven days a week. Count back seven and a half hours: that is the bedtime. Set the alarm for bed.
 - 02 ☐ **Daylight within an hour of waking.** Ten minutes outdoors, without sunglasses, whatever the weather. A window does not count.
 - 03 ☐ **Move for thirty minutes a day,** and three times this fortnight hard enough that talking is difficult. Walking counts. So do stairs.
 - 04 ☐ **Breathe for five minutes** whenever arousal is high or sleep will not come: two inhales through the nose, one long exhale through the mouth, slowly, for five minutes.
 - 05 ☐ **Mark the grid every night,** and on day fifteen write what changed in your decisions, your mood and the sacred hours.
-

WHY IT WORKS Six hours a night produces impairment the impaired cannot detect. Aerobic movement grows the memory-forming brain. Morning light sets the clock that everything else runs on. Slow breathing lowers arousal faster than most alternatives. All four are free.

WHERE IT FAILS Deciding daily. If the four have to be chosen each morning, they will lose to whatever is nearer. Decide once, in writing, and run the fortnight.

LEDGER · INSTRUMENT ELEVEN

The four

WAKE AT _____ BED AT _____ FROM THE _____

SLEEP	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
LIGHT	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
MOVE	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
BREATHE	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Day fifteen: what changed in my decisions.

What changed in my mood, and in the sacred hours.

The one of the four I will keep for the rest of the year, and what it needs.

_____ SIGNED	_____ REVIEWED ON	_____ KEPT / REVISED
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XII

The Others

Good relationships keep us happier and healthier. Period.

*The clearest message that we get from this 75-year study is this:
Good relationships keep us happier and healthier. Period.*

ROBERT WALDINGER, DIRECTOR, HARVARD STUDY OF ADULT
DEVELOPMENT, 2015

IN 1938 RESEARCHERS at Harvard began following two groups of young men: two hundred and sixty-eight sophomores from the college, and four hundred and fifty-six boys from the poorest neighbourhoods of Boston. Seven hundred and twenty-four people, tracked through the war, their marriages, their careers, their illnesses and their deaths, and now their children. It is the longest study of adult life ever conducted, and its current director, Robert Waldinger, was asked, after seventy-five years of data, what it had found. He said it in two sentences, and the second one was *Period*.

What predicted health at eighty was not cholesterol at fifty. It was how satisfied people were, at fifty, with their relationships. The lonely declined earlier, in body and in memory, and the well-connected, whatever their income and whether or not their marriages were peaceful, were protected. Bronnie Ware's dying patients said the same thing from the other end: they wished they had stayed in touch with their friends, and they wished they had said what they felt.

Eleven instruments have been about you. This one is about who you become with, and it is last because the others make it possible: you now have weeks that are counted, asks that are made, and a body that can carry you to people.

— WHY IT WORKS

Relationships behave like the body: they need maintenance and they decay silently. Waldinger and his co-author call it social fitness, and the phrase is exact. Nobody decides to lose a friend. The weeks are simply spent elsewhere, one at a time, and the friendship is discovered to be gone only when it is needed. Robin Dunbar's work on the size of human networks suggests they come in layers of roughly five, fifteen, fifty and a hundred and fifty, and that the inner five, the people you would call at three in the morning, get the largest share of the time we have for others. The number is disputed. The layer is not. Everyone knows who their five are, and almost everyone, asked when they last spoke to each of them properly, is embarrassed by the answer.

One intervention has been tested more than any other. In 2005 Martin Seligman and colleagues randomised people to a set of exercises. In one, part-

icipants wrote a letter of gratitude to someone who had changed their life and never been properly thanked, and delivered it in person, reading it aloud. It produced the largest immediate rise in happiness of any exercise in the study, and the effect was still measurable a month later. A second exercise, writing down three good things that had happened each day and why, was slower to act and lasted six months.

The people who were most satisfied in their relationships at fifty were the healthiest at eighty.

And generosity, which sounds like a cost, turns out to be a strategy. Adam Grant's research on givers, takers and matchers found givers over-represented at the bottom of every success ranking and also, unexpectedly, at the top. The difference between the two groups was not how much they gave but whether they had boundaries. The successful givers gave in ways that did not consume them.

Beyond the five there is the wider ring, and it matters more than it seems. When Gillian Sandstrom and Elizabeth Dunn had people count their daily interactions, those who had more exchanges with acquaintances, the barista, the neighbour, the colleague from another floor, reported belonging more and feeling happier on those days. The weak ties that brought information in the tenth instrument also carry mood. A life with a strong five and no wider ring is protected but narrow; the ring is cheap to maintain and it is where the new people come from.

— HOW TO HOLD IT

Name the five. Beside each name write the date of the last real conversation, and look at the column. Then take one weekly call from the non-negotiable hours: one of the five, half an hour, every week, for the rest of the year. It is the cheapest instrument in the book and, on the study's evidence, the one with the longest reach.

Write the gratitude letter and deliver it in person. Three hundred words, specific, about what the person did and what it changed. Read it to them. It

is uncomfortable for about ninety seconds and then it is one of the best hours either of you will have this year. And repair one thing. There is a conversation you have been postponing; the fourth instrument told you what postponing costs. Have it this month.

On repair, the research is unusually practical. John Gottman, who watched couples in his laboratory for decades, found that the ones who stayed together were not the ones who argued least. They kept a ratio of about five positive exchanges to every negative one, and they answered small bids for attention, a remark about the weather, a hand on the shoulder, by turning toward the person rather than away, most of the time. The couples who later separated turned toward each other about a third of the time. Relationships are not maintained by grand gestures. They are maintained by the small turn, repeated, and it is the cheapest habit in this book.

The eulogy in the first chapter had a speaker. Go back and look at who you chose. They are on the list of five, or they should be, and the whole year has been, in the end, a way of deserving what you hoped they would say.

PRACTICE · INSTRUMENT TWELVE

The gratitude visit

Twenty minutes to name · sixty minutes to write · one visit · begin within the first seven days

- 01 ☐ **Write the five names.** The people whose presence in your eulogy you would most want. Beside each, the date of the last real conversation.
 - 02 ☐ **Write a gratitude letter.** Three hundred words to one person who changed your life and was never properly thanked. Specific: what they did, what it changed.
 - 03 ☐ **Deliver it in person, and read it aloud.** If distance makes that impossible, by call, with the camera on. Write on the ledger when.
 - 04 ☐ **Schedule the weekly call.** One of the five, thirty minutes, same slot each week, for the next twelve weeks. Mark the grid.
 - 05 ☐ **Repair one thing.** The conversation you have been postponing. Have it this month. Write what it was, not what was said.
-

WHY IT WORKS Relationship satisfaction at fifty predicts health at eighty better than any biomarker. The gratitude visit is the fastest-acting exercise in the literature. A weekly call is social fitness made into an appointment, which is the only form in which it survives.

WHERE IT FAILS Sending the letter instead of reading it. The reading is the mechanism. And choosing five people who are all easy: at least one should be someone you have been avoiding.

LEDGER · INSTRUMENT TWELVE

The five

NAMED ON _____

NAME	LAST REAL CONVERSATION	NEXT

The letter: to whom, delivered on.

Twelve weekly calls.

1	2	3	4	5	6	7
8	9	10	11	12		

The repair.

_____ SIGNED	_____ REVIEWED ON	_____ KEPT / REVISED
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The Vow

BENJAMIN FRANKLIN, at twenty, made himself a small book with a page for each of thirteen virtues and seven columns for the days of the week, and every evening he marked with a black spot each fault against the virtue of the week. A course took thirteen weeks; he ran four a year; and he kept the book, on and off, for the rest of his life. He never reached the perfection he was aiming at, he wrote, and was a better and happier man for the attempt. It is the earliest instrument of this kind we know of in English, and the twelve ledgers you have filled are its descendants.

The year has rested on one of the best-replicated findings in psychology. Intentions of the form *I will exercise more* change almost nothing. Intentions of the form *when it is seven on Tuesday, I will put on my shoes and walk to the park* change a great deal: across ninety-four tests and more than eight thousand people, tying an action to a specific situation had a medium-to-large effect, and it held for goals people had already failed at. Peter Gollwitzer calls them implementation intentions. The situation does the remembering, so the person does not have to.

There is an older word for a decision made now that binds you later. Ulysses had himself tied to the mast so that he could hear the sirens without steering towards them; the philosopher Jon Elster called it precommitment. Ulysses did not trust the man he would be in an hour, and he was wise enough to arrange things while he was still the man who could. The kill criterion in the fourth month was a rope of this kind. So is the page opposite.

Write one vow. Not twelve, not a list: one sentence, in the form the research recommends, tying a situation you will certainly meet to an action you have already tested this year, in the name of the person the eulogy described and your ledgers say you have begun to be. Every action, as James Clear puts it, is a vote for the kind of person you wish to become. Sign it. If a witness is available, have them sign it too. Then close the book, and open the envelope.

THE INSTRUMENT

The Vow

Having used the twelve instruments over the course of one year, I make this vow, in the form of an implementation intention, and enter it into the record of this copy.

I, _____, on _____,

when _____

_____,

will _____

_____,

because I am the kind of person who

_____.

To be read again on the first day of each of the next four quarters, and marked, on each occasion, kept or revised.

QUARTER ONE

QUARTER TWO

QUARTER THREE

QUARTER FOUR

SIGNED

WITNESSED

COPY NUMBER

The Twelve Axioms

- I THE EULOGY *Begin at the end.*
- II THE WEEKS *A life is four thousand weeks. Spend them on purpose.*
- III THE QUESTION *The quality of a life follows the quality of its questions.*
- IV INVERSION *Avoid stupidity before you attempt brilliance.*
- V ASYMMETRY *Make many small bets whose downside is bounded and whose upside is not.*
- VI THE LOOP *You become whatever you get fast feedback on.*
- VII ATTENTION *Your experience is what you agree to attend to.*
- VIII COMPOUNDING *Small, repeated, for years, beats large, occasional, for months.*
- IX TASTE *Quantity is the road to quality.*
- X THE ASK *Most of the world is a door that opens if you knock.*
- XI THE BODY *The mind is something the body does.*
- XII THE OTHERS *Good relationships keep us happier and healthier. Period.*

Notes & Sources

Where a chapter rests on research, the research is named here, with what it shows and what it does not. Effect sizes are given where they matter. A study of undergraduates recalling Swahili is evidence about memory, not a promise about your career; the book has tried to claim only what the evidence supports and to label a story as a story.

BEFORE YOU BEGIN

Andy Matuschak, “Why books don’t work” (2019), andymatuschak.org; the argument that books rest on an unstated model of learning, transmissionism, and that most readers retain little and rarely notice. On price and use: John Gourville and Dilip Soman, “Pricing and the Psychology of Consumption”, *Harvard Business Review*, September 2002, on health-club members whose attendance tracked how recently they had paid. Baba Shiv, Ziv Carmon and Dan Ariely, *Journal of Marketing Research* 42 (2005): people who paid a discounted price for an energy drink performed worse on puzzles than those who paid full price. The claim that an expensive book gets used is a claim about tendency, not a guarantee.

I · THE EULOGY

Bronnie Ware, *The Top Five Regrets of the Dying* (Balboa Press, 2011; Hay House, 2012), from her 2009 essay “Regrets of the Dying”; the five regrets are paraphrased in her order. Daniel H. Pink, *The Power of Regret* (Riverhead, 2022): the World Regret Survey (more than 16,000 regrets from 105 countries at publication) and the American Regret Project (4,489 adults); the four core regrets; the finding that inaction regrets overtake action regrets with age, about two to one overall, building on Thomas Gilovich and Victoria Medvec, *Psychological Review* 102 (1995). Jeff Bezos, interview with the Academy of Achievement, 4 May 2001, on the “regret minimization framework”. Hal Hershfield et al., “Increasing Saving Behavior Through Age-Progressed Renderings of the Future Self”, *Journal of Marketing Research* 48 (2011): participants who saw an aged image of themselves allocated \$172 of a hypothetical \$1,000 to retirement, against \$80 for those who saw their present face. Prospective hindsight: see chapter IV. Steve Jobs, Stanford commencement address, 12 June 2005. The Nobel obituary story is widely repeated and unverified; no copy of the newspaper has been located, and it is presented as legend.

II · THE WEEKS

Oliver Burkeman, *Four Thousand Weeks* (Farrar, Straus and Giroux, 2021): “assuming you live to be eighty, you’ll have had about four thousand weeks.” The chart has ninety rows of fifty-two, with the eightieth year marked; eighty years of fifty-two weeks is 4,160, and counting days gives about 4,170. Tim Urban, “Your Life in Weeks” (2014) and “The Tail End” (2015), *Wait But Why*; the parental arithmetic in the chapter follows his method with illustrative numbers.

Seneca, *De Brevitate Vitae* 1.3, in C. D. N. Costa's translation (Penguin, 1997). Annie Dillard, *The Writing Life* (Harper & Row, 1989), p. 32. Marcus Aurelius, *Meditations* 2.11, in Gregory Hays's translation (Modern Library, 2002); the Greek title, *Ta eis heauton*, means "to himself".

III · THE QUESTION

Richard Hamming, "You and Your Research", Bell Communications Research colloquium, 7 March 1986. Jacob Getzels and Mihaly Csikszentmihalyi, *The Creative Vision: A Longitudinal Study of Problem Finding in Art* (Wiley, 1976): art students observed choosing and arranging objects before drawing; those who explored longer produced work rated more original, and problem-finding behaviour predicted professional success in follow-ups years later. The sample was small and the field was art; the chapter treats the result as suggestive, not settled. Rainer Maria Rilke, *Letters to a Young Poet*, letter of 16 July 1903, translated by M. D. Herter Norton (Norton, 1934). The "five whys": Taiichi Ohno, *Toyota Production System* (1978; English 1988). Paul Graham, "How to Do Great Work" (July 2023), paulgraham.com. "What would have to be true" is Roger Martin's formulation, in *Playing to Win* (with A. G. Lafley, Harvard Business Review Press, 2013). The term *eigenquestion* is used by Shishir Mehrotra. Leonardo's lists of questions are in the Codex Atlanticus and are discussed in Walter Isaacson, *Leonardo da Vinci* (Simon & Schuster, 2017).

IV · INVERSION

Charles T. Munger, "Prescriptions for Guaranteed Misery in Life", commencement address, Harvard School, Los Angeles, 13 June 1986; the German maxim is traditionally attributed to Carl Gustav Jacob Jacobi but has no located primary source. Deborah Mitchell, J. Edward Russo and Nancy Pennington, "Back to the Future: Temporal Perspective in the Explanation of Events", *Journal of Behavioral Decision Making* 2 (1989): explaining an outcome as certain and past produced about thirty per cent more reasons, and more specific ones; the study measured the number and specificity of reasons, not their accuracy. Gary Klein, "Performing a Project Premortem", *Harvard Business Review*, September 2007. Roger Buehler, Dale Griffin and Michael Ross, "Exploring the 'Planning Fallacy'", *Journal of Personality and Social Psychology* 67 (1994): thirty-seven honours students predicted 33.9 days for their theses (48.6 in the worst case) and took 55.5. Daniel Kahneman and Amos Tversky named the planning fallacy in 1979. Annie Duke, *Quit* (Portfolio, 2022), on kill criteria as "states and dates". The phrase "anti-goals" is Andrew Wilkinson's. The 1935 crash of the Boeing Model 299 at Wright Field and the origin of the pilot's checklist are recounted in Atul Gawande, *The Checklist Manifesto* (Metropolitan, 2009). Alex Haynes et al., "A Surgical Safety Checklist to Reduce Morbidity and Mortality in a Global Population", *New England Journal of Medicine* 360 (2009): complications fell from 11.0 to 7.0 per cent and deaths from 1.5 to 0.8 per cent across eight hospitals.

V · ASYMMETRY

Jeff Bezos, 2015 Letter to Shareholders, Amazon.com (April 2016), on Type 1 and Type 2 decisions. Dean Keith Simonton, “Creative Productivity: A Predictive and Explanatory Model of Career Trajectories and Landmarks”, *Psychological Review* 104 (1997), and *Scientific Genius* (Cambridge, 1988): the equal-odds rule holds approximately, and later work (Forthmann et al., 2019) finds departures from it. The bounded-and-unbounded framing, the barbell and the idea of ruin as an absorbing state are Nassim Nicholas Taleb’s, in *Antifragile* (Random House, 2012) and *Skin in the Game* (2018). Annie Duke, *Thinking in Bets* (Portfolio, 2018), on “re-sulting”. Peter F. Drucker, “Managing Oneself”, *Harvard Business Review*, March–April 1999: “Whenever you make a key decision or take a key action, write down what you expect will happen. Nine or 12 months later, compare the actual results with your expectations.” Benjamin Franklin to Joseph Priestley, London, 19 September 1772, on “Moral or Prudential Algebra”. J. L. Kelly, “A New Interpretation of Information Rate”, *Bell System Technical Journal* 35 (1956); Edward O. Thorp, *A Man for All Markets* (Random House, 2017).

VI · THE LOOP

K. Anders Ericsson, Ralf Krampe and Clemens Tesch-Römer, “The Role of Deliberate Practice in the Acquisition of Expert Performance”, *Psychological Review* 100 (1993); Ericsson’s correction of the “10,000-hour rule” is in Ericsson and Robert Pool, *Peak* (Houghton Mifflin Harcourt, 2016). Brooke Macnamara, David Hambrick and Frederick Oswald, “Deliberate Practice and Performance in Music, Games, Sports, Education, and Professions: A Meta-Analysis”, *Psychological Science* 25 (2014), eighty-eight studies: 26, 21, 18, 4 and under 1 per cent of variance explained respectively. The reading of the professional figure as a loop-length problem is the book’s, not the authors’. Jeffrey Karpicke and Henry Roediger III, “The Critical Importance of Retrieval for Learning”, *Science* 319 (2008): about 80 per cent recall after a week with repeated retrieval against about 36 per cent with repeated study. Nicholas Cepeda et al., *Psychological Bulletin* 132 (2006) and *Psychological Science* 19 (2008), on spacing. Philip Tetlock and Dan Gardner, *Superforecasting* (Crown, 2015); the “30 per cent better than intelligence analysts” figure was reported by David Ignatius, *Washington Post*, 1 November 2013, and is cited by Tetlock. Charles Goodhart’s observation dates from 1975 and was generalised by Marilyn Strathern in 1997. Richard Feynman, “Cargo Cult Science”, Caltech commencement, 1974. Atul Gawande, “Personal Best”, *The New Yorker*, 3 October 2011.

VII · ATTENTION

William James, *The Principles of Psychology* (Henry Holt, 1890), vol. 1, ch. XI, p. 402. Herbert Simon, “Designing Organizations for an Information-Rich World” (1971). Sophie Leroy, “Why Is It So Hard to Do My Work? The Challenge of Attention Residue When Switching Between Work Tasks”, *Organizational Behavior and Human Decision Processes* 109 (2009). Adrian Ward, Kristen Duke, Ayelet Gneezy and Maarten Bos, “Brain Drain: The Mere Pres-

tion for Consumer Research 2 (2017). Benjamin Baird et al., “Inspired by Distraction: Mind Wandering Facilitates Creative Incubation”, *Psychological Science* 23 (2012). Cal Newport, *Deep Work* (Grand Central, 2016). The four-hour limit and afternoon naps are reported in Ericsson et al. (1993), above. Simone Weil, letter to Joë Bousquet, 13 April 1942. A waking day of a thousand minutes assumes sixteen and a half to seventeen hours awake. The touch counts are from dscout, “Mobile Touches” (2016): 2,617 touches a day on average, 5,427 for the top decile of users.

VIII • COMPOUNDING

James Clear, *Atomic Habits* (Avery, 2018): $1.01^{365} = 37.78$. The table in the chapter uses weekly rates compounded over 52-week years; the figures are rounded. Hershfield et al. (2011), see chapter I. Cal Newport, *So Good They Can't Ignore You* (Business Plus, 2012), on career capital. Naval Ravikant, “How to Get Rich (without getting lucky)”, 31 May 2018: leverage from “capital, people, and products with no marginal cost of replication (code and media)”, the last two being “permissionless leverage”; the epigraph is from the same thread. The rule about never interrupting compounding unnecessarily is widely attributed to Charles Munger. The claim that permissionless leverage has grown by an order of magnitude since 2018 is the book’s own estimate. Buffett’s net worth by age is documented in public filings and in Alice Schroeder, *The Snowball* (Bantam, 2008). The Lindy effect is discussed in Nassim Nicholas Taleb, *Antifragile*.

IX • TASTE

David Bayles and Ted Orland, *Art & Fear* (Capra Press, 1993): the ceramics story is presented there as an anecdote; Orland later said it came from Jerry Uelsmann’s photography class at the University of Florida, with the medium changed. Simonton (1997), see chapter V. Ira Glass, “Ira Glass on Storytelling”, part 3, Current TV (2009); the wording follows the standard transcript. Henri Poincaré, *Science et Méthode* (1908), translated by Francis Maitland (1914). John Locke, “A New Method of Making Common-Place-Books” (London, 1706; first published in French, 1686). Marcus Aurelius, see chapter II. The “first thirty are everybody’s ideas” heuristic and the one-to-one ratio are practice, not research. Montaigne’s inscribed library beams are described in Sarah Bakewell, *How to Live* (Chatto & Windus, 2010).

X • THE ASK

Richard Wiseman, *The Luck Factor* (Century, 2003) and “The Luck Factor”, *Skeptical Inquirer* 27 (2003): the newspaper exercise is a demonstration reported by the author, not a controlled trial. Francis Flynn and Vanessa Lake (now Bohns), “If You Need Help, Just Ask: Underestimating Compliance with Direct Requests for Help”, *Journal of Personality and Social Psychology* 95 (2008): predicted numbers of people to approach were roughly double the actual numbers. Mark Granovetter, “The Strength of Weak Ties”, *American Journal of Sociology* 78 (1973): of fifty-four respondents who found jobs through a contact, 16.7 per cent saw the contact often, 55.6 per cent occasionally and 27.8 per cent rarely. Jon Jecker and David Landy,

“Liking a Person as a Function of Doing Him a Favour”, *Human Relations* 22 (1969); Franklin’s account is in his *Autobiography*. Jia Jiang, *Rejection Proof* (Harmony, 2015). Steve Jobs told the Hewlett story in his 1994 interview with the Silicon Valley Historical Association: “Most people never pick up the phone and call. Most people never ask.”

XI · THE BODY

Hans Van Dongen, Greg Maislin, Janet Mullington and David Dinges, “The Cumulative Cost of Additional Wakefulness”, *Sleep* 26 (2003): forty-eight adults, fourteen days at four, six or eight hours in bed; six hours or less produced deficits “equivalent to up to 2 nights of total sleep deprivation”, and subjects were “largely unaware of these increasing cognitive deficits”. Kirk Erickson et al., “Exercise Training Increases Size of Hippocampus and Improves Memory”, *PNAS* 108 (2011): 120 adults aged 55–80; the walking group’s anterior hippocampus grew about 2 per cent over a year while the stretching group’s shrank about 1.4 per cent. Charles Hillman, Kirk Erickson and Arthur Kramer, “Be Smart, Exercise Your Heart”, *Nature Reviews Neuroscience* 9 (2008). Melis Yilmaz Balban et al., “Brief Structured Respiration Practices Enhance Mood and Reduce Physiological Arousal”, *Cell Reports Medicine* 4 (2023): 108 participants completed a month of five daily minutes; cyclic sighing improved positive affect more than mindfulness meditation. The morning-light prescription follows the consensus of circadian research rather than a single trial. Marilyn Oppizzo and Daniel Schwartz, “Give Your Ideas Some Legs”, *Journal of Experimental Psychology: Learning, Memory, and Cognition* 40 (2014): creative output rose by an average of 60 per cent while walking. None of this is medical advice.

XII · THE OTHERS

The Harvard Study of Adult Development began in 1938 with 268 Harvard sophomores and 456 boys from Boston’s poorest neighbourhoods. Robert Waldinger, “What Makes a Good Life?”, TEDxBeaconStreet, November 2015; Robert Waldinger and Marc Schulz, *The Good Life* (Simon & Schuster, 2023), on “social fitness”. Robin Dunbar, *Journal of Human Evolution* 22 (1992) and *Friends* (Little, Brown, 2021); a 2021 reanalysis by Lindenfors, Wartel and Lind found the confidence interval around 150 too wide to support a specific number, which the chapter notes. Martin Seligman, Tracy Steen, Nansook Park and Christopher Peterson, “Positive Psychology Progress: Empirical Validation of Interventions”, *American Psychologist* 60 (2005): the gratitude visit produced large positive change for one month; “three good things” increased happiness and decreased depressive symptoms for six months. Adam Grant, *Give and Take* (Viking, 2013). Gillian Sandstrom and Elizabeth Dunn, “Social Interactions and Well-Being: The Surprising Power of Weak Ties”, *Personality and Social Psychology Bulletin* 40 (2014). John Gottman, *Why Marriages Succeed or Fail* (Simon & Schuster, 1994) and, with Joan DeClaire, *The Relationship Cure* (Crown, 2001), on the five-to-one ratio and on turning toward bids. Bronnie Ware, see chapter I.

THE VOW

Benjamin Franklin, *Autobiography*, part two: thirteen virtues, a page per virtue, seven columns, a black spot per fault, one virtue a week, four courses a year. Peter Gollwitzer and Paschal Sheeran, “Implementation Intentions and Goal Achievement: A Meta-Analysis of Effects and Processes”, *Advances in Experimental Social Psychology* 38 (2006): ninety-four independent tests, more than eight thousand participants, $d = 0.65$. Jon Elster, *Ulysses and the Sirens* (Cambridge, 1979). James Clear, *Atomic Habits*, ch. 2. Aristotle, *Nicomachean Ethics*, book II. Two popular ideas are absent from this book on purpose: “grit” and “growth mindset”. Meta-analyses by Credé, Tynan and Harms (2017) and Sisk et al. (2018) found the first largely indistinguishable from conscientiousness and the second’s interventions to have very small effects. They may yet prove useful. The evidence is not strong enough to build a month on.

Colophon

THE INSTRUMENT was written, designed and typeset in 2026. The text is set in EB Garamond, Georg Duffner's revival of Claude Garamond's sixteenth-century romans, with Instrument Serif for display and Instrument Sans for the working pages, both by Rodrigo Fuenzalida and Jordan Egstad. All three are released under the SIL Open Font License.

Printed in two colours, black and cinnabar, on Munken Pure Rough 120 gsm, an uncoated paper chosen because it takes a fountain pen. Sewn in sixteen-page signatures with a hollow back so that the book opens flat for writing. Bound in cloth over boards, blocked in copper foil, with printed endpapers, two ribbon markers and a cloth clamshell box. The Twelve Cards were printed letterpress from photopolymer plates on 100% cotton stock. The Weeks Chart was printed offset and folded by hand.

The edition is limited to five hundred numbered copies and twelve *hors commerce*. It will not be reprinted in this form.

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